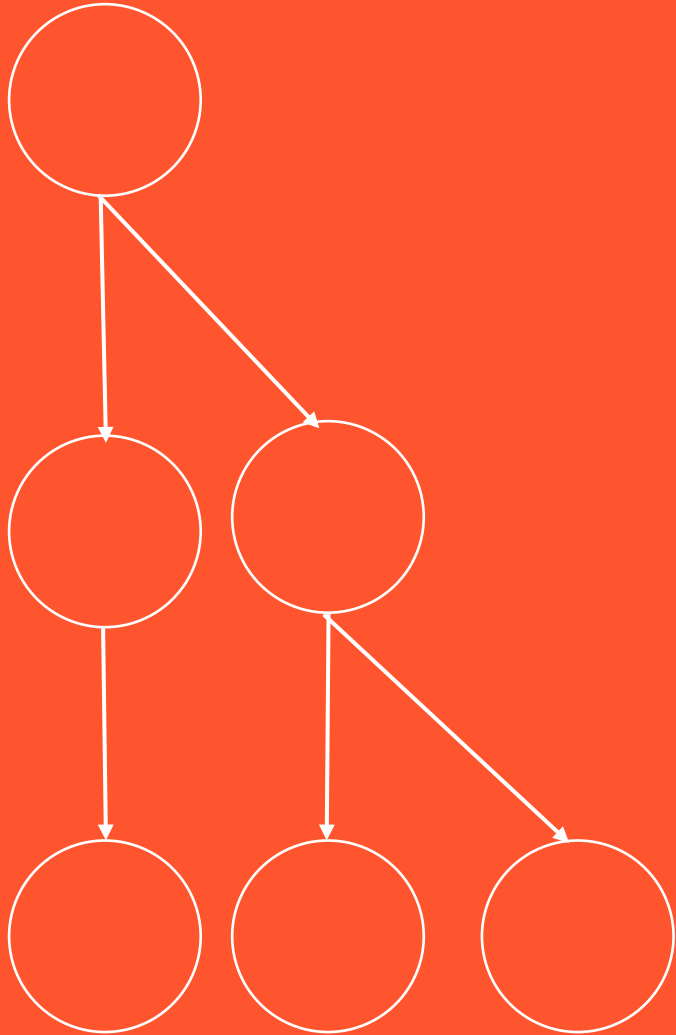


BCOG 100

The Evolution of Neurons

Lecture 1: Evolution and the Beginning of Life



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 4-1

Password: neuron

Question: Neurons are described as creating “electrochemical” communication. What does that mean?



From Evolution to Neurons

Today's Outline

- Why biology in cognitive science?
- Evolution by natural selection
- Origins of life & early “intelligence”



From Evolution to Neurons

Why Biology Matters for Cognitive Science

- Brains are biological organs
- Evolution shaped cognition by shaping the brain
- Constraints, tradeoffs, and possibilities
- Simple life and evolution itself can be thought of as a kind of intelligent, information, processing, and algorithm

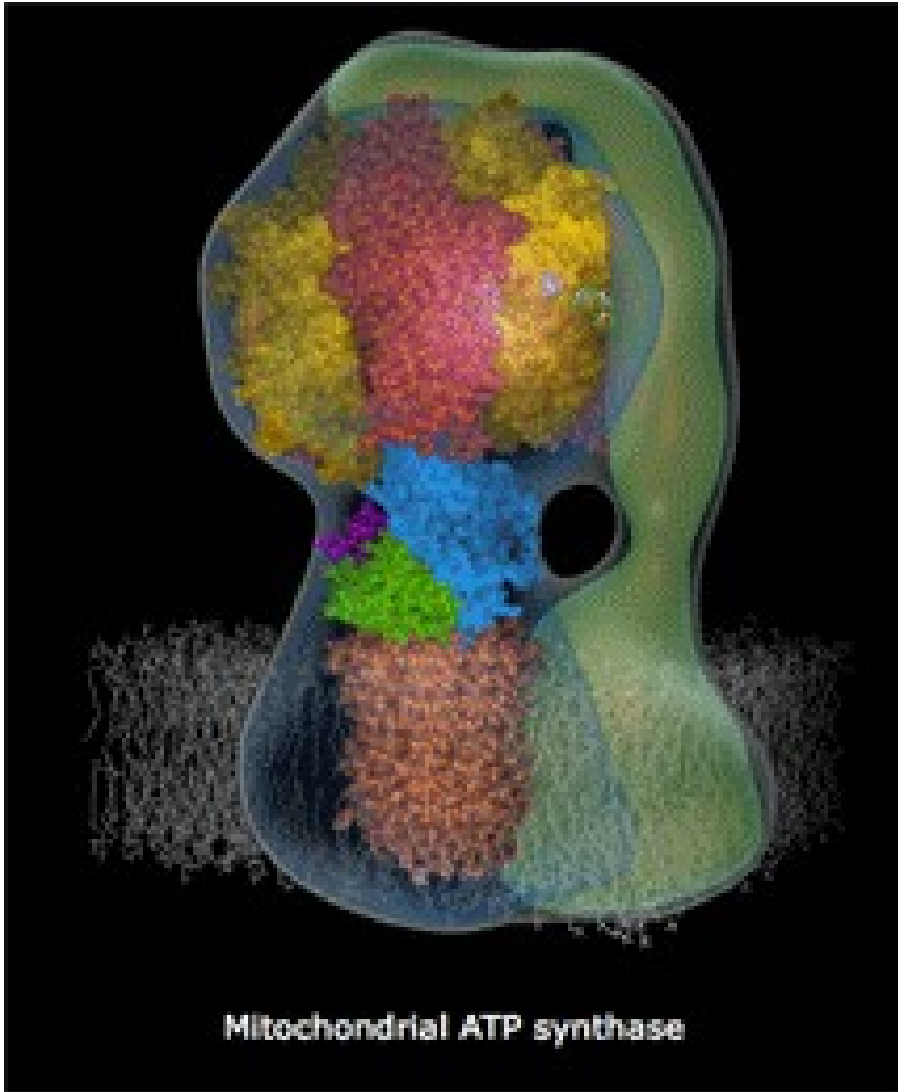
Over the next 8 weeks we'll go from:

- Molecules → Cells → Neurons
- Circuits
- Brains
- Higher Cognitive Abilities like Symbolic Reasoning, Language and Social Cognition.

This mirrors both evolution *and* our course.



The Energy Challenge in Early Life



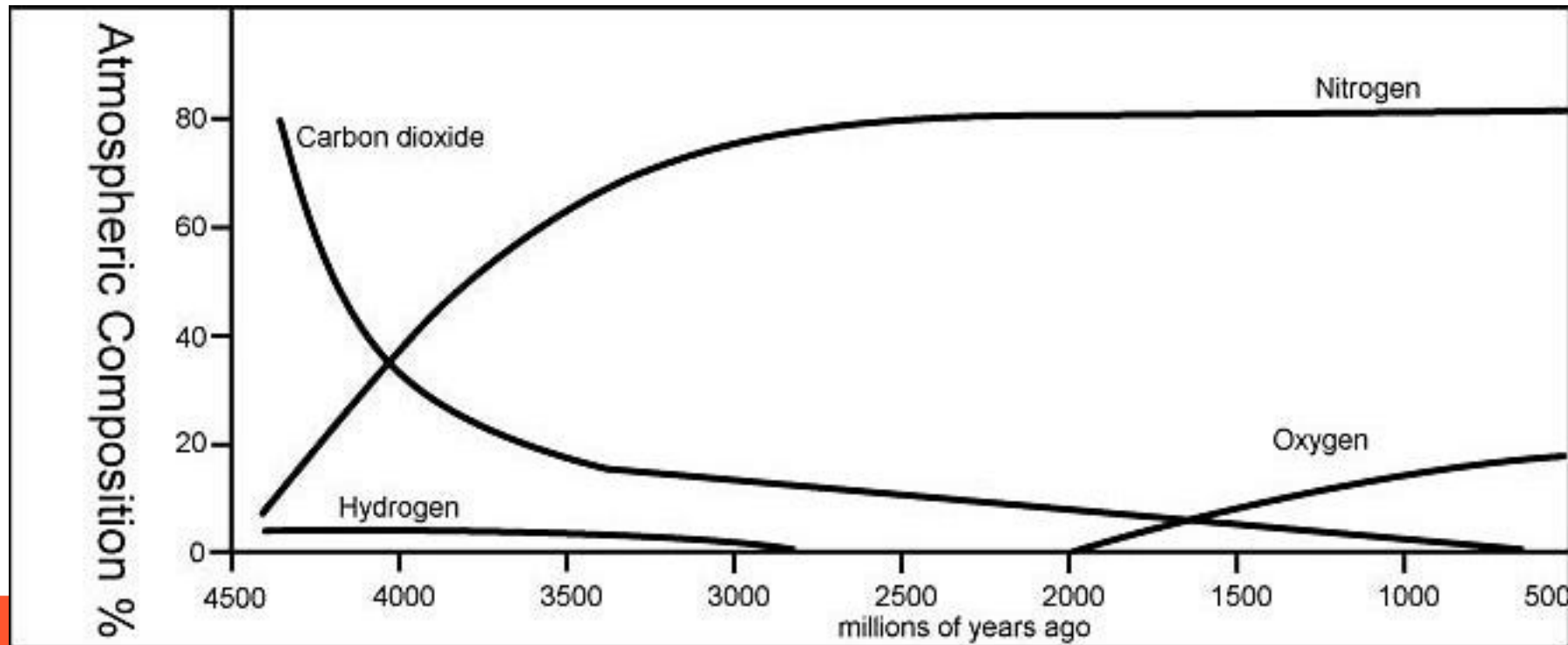
The Energy Challenge

- ATP: adenosine triphosphate, a cell's energy currency
- Early life: hydrogen fuel $\rightarrow$ ATP (inefficient)
- Photosynthesis: sunlight + $\text{CO}_2 \rightarrow$ sugar $\rightarrow$ ATP (efficient, renewable)

The Energy Challenge in Early Life

Great Oxygenation Event

- Photosynthesis released oxygen (O_2)
- Oxygen = toxic & reactive → mass extinction
- Oxygen = powerful electron acceptor
- Respiration with O_2 → ~15× more ATP



Summarizing Evolution of Life Before Neurons

Today we saw:

- Evolution by natural selection = variation + inheritance + selection
- Life from non-life: the RNA world → DNA, proteins, membranes
- Early “intelligence” in single cells (chemotaxis)
- Energy revolution: photosynthesis & oxygen

Next time...

- Oxygen killed many forms of life...
- ...but also powered new complexity
- With this energy, life could afford bigger bodies, new strategies, and the first neurons

